Experiment 1(c): 8-Bit Subtraction with Borrow Using Direct Addressing

AIM:

To write an assembly language program to subtract two 8-bit numbers stored in memory locations 4200H and 4201H and store the result in 4202H and borrow in 4203H using direct addressing.

APPARATUS REQUIRED:

1. 8085 Microprocessor kit – 1
2. Power card – 1
3. Keyboard – 1

ALGORITHM:

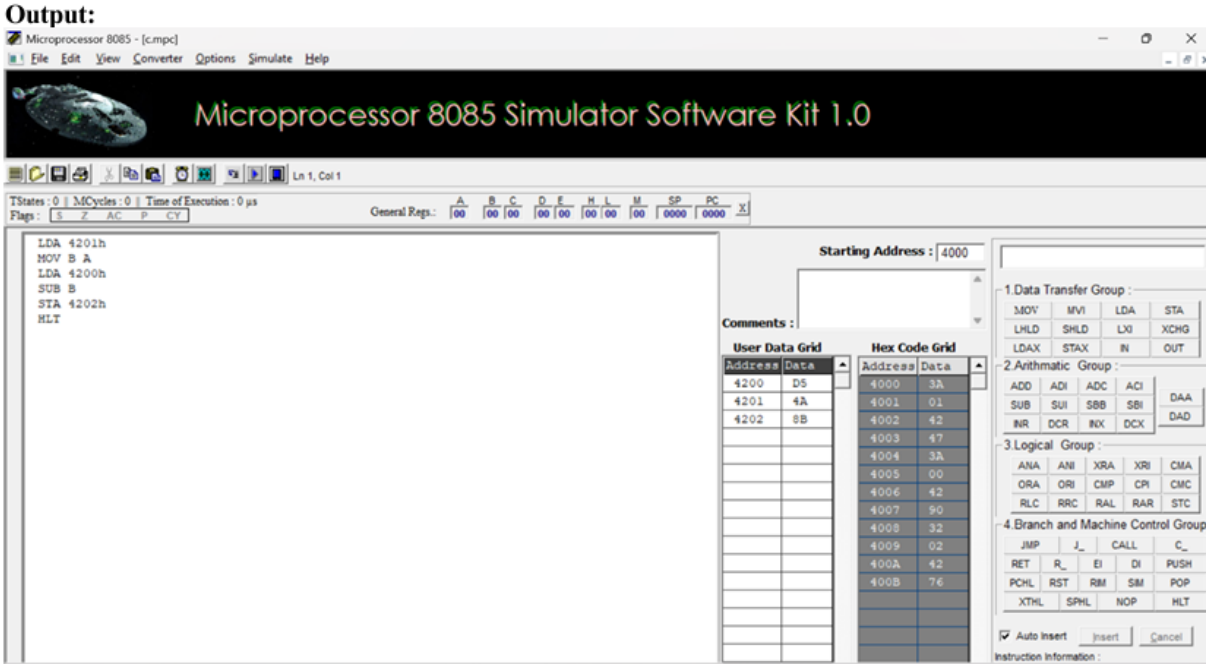
1. Load the second number from memory into the accumulator and move it to register B.
2. Load the first number into the accumulator.
3. Subtract the content of register B from the accumulator.
4. Check the carry flag to determine whether a borrow occurred.
5. Store the difference in memory location 4202H.
6. Store the borrow in memory location 4203H.
7. Stop the program.

PROGRAM:

```
LDA 4200H
MOV B,A
LDA 4201H
SUB B
JNC SKIP
MVI A,01H
STA 4203H
HLT
SKIP: MVI A,00H
STA 4203H
HLT
```

OUTPUT:

Output:



RESULT:

Thus, an assembly language program for subtracting two 8-bit numbers with borrow using direct addressing was written and executed successfully using the 8085 microprocessor.